Banking Malware Containment

Live Demo Script — what to click and what to say, in order

Setup (before the professor looks over your shoulder)

Do

- Open PowerShell in **Desktop\PROJECTS\CNPROJECT**.
- Run **streamlit run app.py** and press Enter.
- A browser tab opens at http://localhost:8501. Maximize it on the projector.

Step 1 — Set the scene

Say

- Banking networks connect workstations, ATMs, branch servers, and the core-banking system. If one machine is infected by a self-propagating worm, it spreads sideways across those connections. The hard problem is not detection — it is deciding which few machines to isolate first when the team can only act on a small number at once.
- This prototype simulates that and produces an ordered quarantine plan under a fixed budget.

Step 2 — Introduce the controls

Do

- Point to the left sidebar.

Say

- On the left I choose the topology — an 8-node branch or a 19-node multi-branch bank — the infected seed machine, the transmission probability, the quarantine budget, and how many trials to average over. Every change recomputes everything live.

Step 3 — Headline result: final infection counts

Do

- Point to the top table “Final infection counts (single run)”.

Say

- With no containment, 8 machines end up infected. The proposed method cuts that to 4 — matching the best baseline and beating random and earliest-first.

Step 4 — Reproduce the report numbers (default scenario)

Do

- In the sidebar set: Seed node = **WS_1**, transmission probability = **0.25**, budget = **2**, Random seed = **42**. Trials can stay 30, or drop to 10 to make slider changes faster.

Say

- These are the same settings as the checked-in experiment, so the numbers match the report.

Step 5 — The strong result: switch to multi-branch

Do

- Set Topology = **Multi-branch (19 nodes)**, Seed node = **WS_N1**, transmission probability = **0.40**, budget = **2**.

Say

- On the bigger network — three branches, six ATMs, a core server — the difference is much clearer.

Step 6 — Mean infections across trials

Do

• Point to the bar chart “Mean final infections across trials”.

Say

• Averaged over 30 runs with 95% confidence intervals: no containment is about 9.5 infected, the proposed method about 4.3 — roughly a 55% reduction, and the lowest of all five strategies.

Step 7 — Infection curves

Do

• Point to the “Infection curves” chart.

Say

• This is spread over time. The proposed strategy's curve stays consistently below no containment.

Step 8 — The network topology

Do

• Point to the “Network topology” graph.

Say

• Here is the actual bank network. Blue is susceptible, red is infected, orange is quarantined. The plan quarantines the core server and a workstation, cutting the main lateral path.

Step 9 — The quarantine plan (the deliverable)

Do

• Point to the “Quarantine plans” table.

Say

• This is the actionable output — a ranked, budget-limited list of machines to isolate, in order. Not “quarantine everything”, but a specific plan for the response team.

Step 10 — Live interaction (the money shot)

Do

• Drag the transmission-probability slider up, then drag the budget slider up.

Say

• Watch — if I make the worm more aggressive, infections rise. If I give the team a bigger budget, infections drop. Everything recomputes live. This is the real simulation, not a recording.

Step 11 — Wrap up

Say

• So: a budget-aware, topology-informed containment method. It is a simulation — it never touches a real network — but it demonstrates the decision logic a response team would use.

If something goes wrong

Do

• If the dashboard will not load, run the CLI instead: **python -m src.comparison --config config/multibranch.json**.

• It prints the same numbers and writes charts to results/multibranch/. Open those PNGs and talk through the same points.

Two tips

Do

- Change one thing at a time and narrate the change.
- If slider changes feel slow, set Trials to 10.